From angular structure to conditional kinematic constraints

A converged baseline from radiation, astrometry, optical distances and finite geometry images

28 October 2026

Physical question and common representation

Angular structure becomes a constraint on spacetime only after its response to a specified physical state is identified. A temperature multipole, an apparent-magnitude quadrupole and an astrometric electric quadrupole are different observables. Their tensor representations can preserve orientation and sign, but their physical meanings follow from radiative transfer, optical propagation and source motion. The common inference problem is therefore

$$Y = \mathcal{R}_\eta(X_{\text{phys}}) + \epsilon, \qquad \mathcal{K}(Y) = \{\theta : \mathcal{R}_\eta(\theta) \text{ is compatible with } Y \text{ for an admitted } \eta\}.$$

Here η contains physical histories, processing and calibration variables appropriate to the observation. A compatibility region can constrain a named combination even when the individual parameters remain degenerate. The present results connect retained observations and finite response calculations to four targets: radiation-to-kinematics, beyond-FLRW kinematic quantities, physical-family classification, and direction/redshift-dependent expansion.

The five coordinates (x, F, Q, Π, G_F) organize these calculations without changing the observational law. The carrier x retains signed coefficients, their representation, frame and units. For a physical image $\mathcal{C}$, $F(W) = \sup_{x\in\mathcal{C}}\langle W, x\rangle$ measures its extent in a declared direction. A gauge Q measures displacement relative to a specified centred reference body; a positive cone or an uncentred trace-one slice is not automatically a usable gauge body. Π records probability only under the declared sampling or posterior law. G_F retains the comparison maps and cross-window dependence needed to follow the same object through depth or transport. Nonconvex family labels remain attached to their images because a support function alone records their convex hull.

This distinction is operational. In the saved Gaussian-carrier comparison, tensorized and correctly specified joint GLS estimates agreed to a maximum absolute difference of 2.42×10^{-13}, and their information matrices to 2.15×10^{-16}. A common response-family carrier had dimension 12, whereas a single-family carrier had dimension 6. The former retained the compared likelihood ratios; the latter could erase their discrimination. These are fixed-law compression results, with the physical response held constant [E1].

A supplied approximation bound can be transported through the same structure. If $e^T G e \le \epsilon_X^2$ and its observation response is $b = Be$, then

$$\sup_e v^T B e = \epsilon_X \sqrt{v^T B G^{-1} B^T v}.$$

The matrix in this expression describes a deterministic joint error body. Keeping its shared directions can be substantially sharper than bounding each observable separately. After nuisance

projection, this body induces a bound on parameter bias and a corresponding confidence region. In a finite five-parameter Gaussian control with bias radius 1.5 in Fisher units, the ordinary nominal 95% ball had exact coverage 82.7209%; the bias-aware construction had exact coverage 95% and Monte Carlo coverage 95.098% in 100,000 draws [E2]. The physical significance is that truncation or response error enters the inferential region instead of disappearing behind a small statistical error bar.

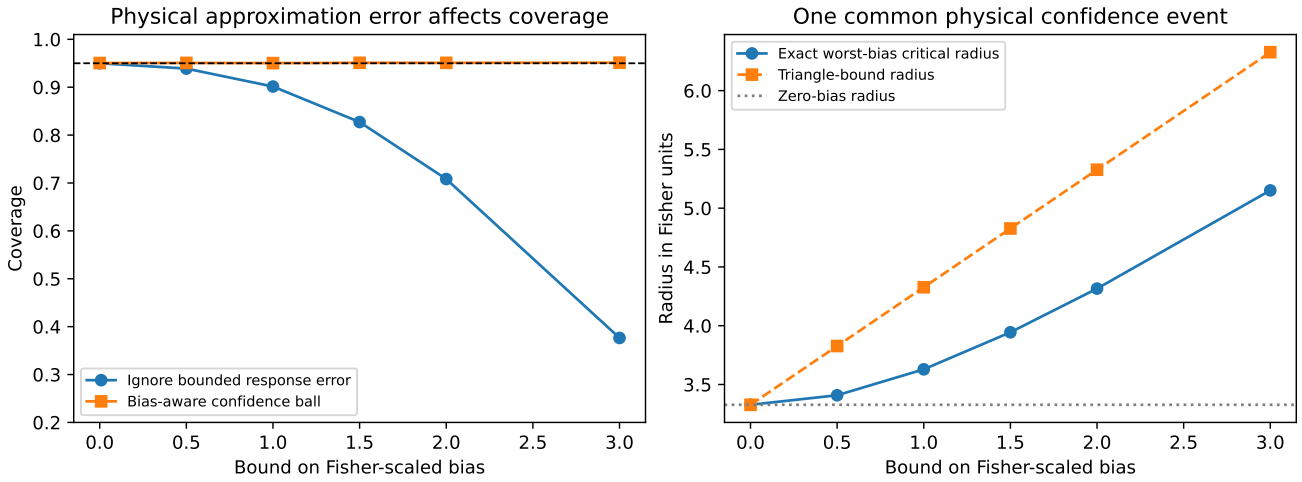


Figure 1: A common physical error bound propagated to parameter coverage, compared with ignoring the bounded bias.

Internal polarization illustrates the same distinction at the detector level. The positive distribution $f = f_0[I + (n \cdot e)\theta \cdot \sigma]$, $\|\theta\| \leq 1$, can preserve total trace and stress while retaining opposite channel currents. In the executed finite count experiment, calibrated channel data distinguished two opposite-sign states with 98.086% accuracy in 100,000 equal-prior trials; trace or sorted state spectra gave 50%. Known zero calibration allowed three pooled sufficient counts, whereas an unknown common calibration required the six-channel natural-parameter span [E12]. This is an information statement for the specified count law. A geometric helicity phase connection and a physical polarization converter have different measurement spans and are kept distinct.

1. Low-multipole morphology and the radiation-to-shear response

The local radiation field carries an energy density, flux and anisotropic stress. For a complete blackbody temperature sky the bolometric intensity is proportional to T^4, so a linear temperature-to-stress conversion need not preserve the finite physical target. This has already been tested using the same positive synthetic skies and correlated temperature errors for both adapters. In the isotropic-rest-frame case with observer speed $\beta = 0.00123$, the exact adapter's radiative tensor target had 94.531% coverage, while the linear adapter had 74.097%. The scalar rest-frame stress-anisotropy target had 95.557% versus 37.720% coverage. At $\beta = 0.3$, the corresponding exact coverages were 96.045% and 95.386%, while both linear physical-target coverages were zero. Each case used 8,191 calibration skies and 4,096 independent test skies [E3].

The tensor target is the finite-boost stress residual

$$\mathcal{D} = \frac{\pi}{\rho} - k_\gamma(f \cdot f)\,\mathrm{STF}(f \otimes f), \qquad f = \frac{q}{c\rho}, \qquad k_\gamma(s) = \frac{3}{2 + \sqrt{4 - 3s}}.$$

It tests whether the measured energy flux and anisotropic stress have the relation induced by a boosted isotropic radiation stress tensor. The companion scalar is the norm of the Landau-frame

pressure departures from $\rho_L/3$, divided by ρ_L. These stress-level targets distinguish observer-induced dipole structure from a residual rest-frame anisotropic stress; they do not replace the higher-multipole distribution or the spacetime response.

The linear method remained calibrated for its own approximate target. Its loss concerned the physical quantity to which that target was being assigned. This provides a concrete reason to retain the physical response map through inference. It also separates a response bias from a failure of a numerical likelihood algorithm.

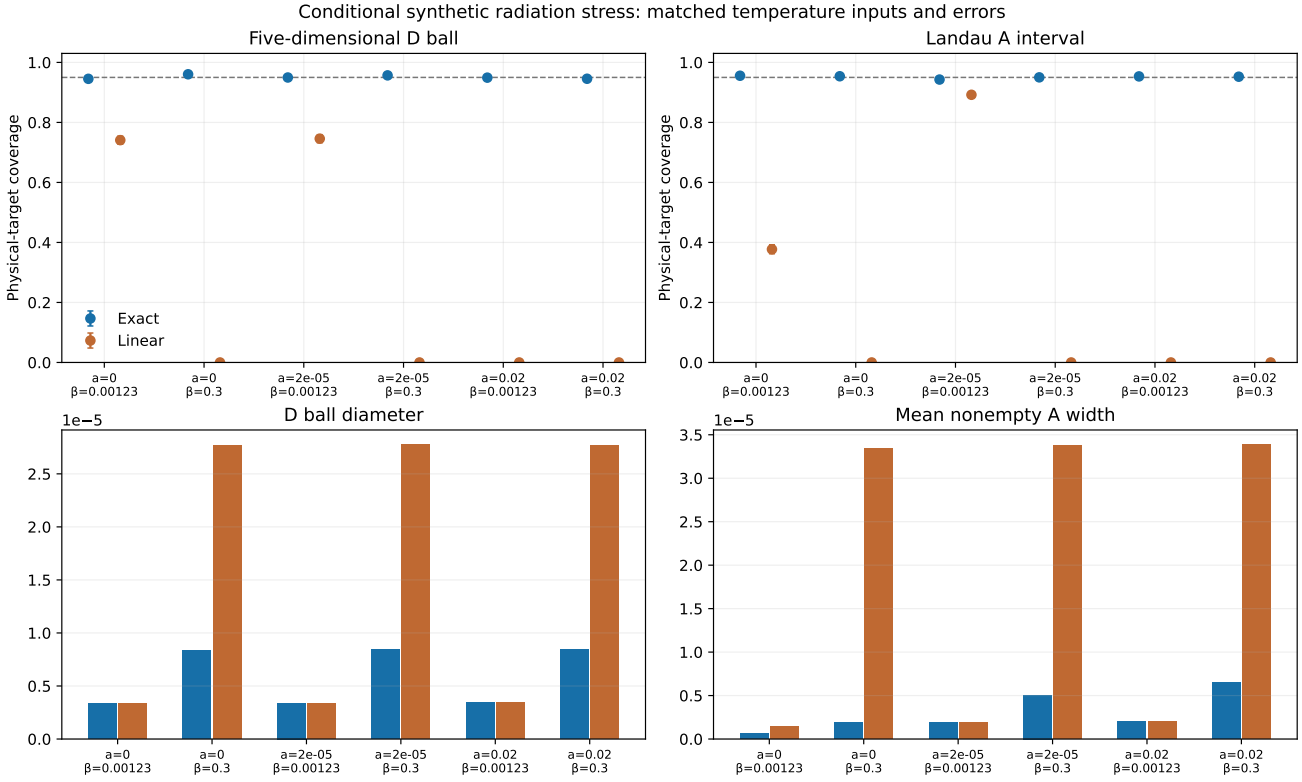


Figure 2: Common-input radiation comparison: exact and linear temperature-to-stress adapters under the same finite synthetic law.

The instantaneous shear-response operator $\mathcal{L}_I$ acts on the five-dimensional STF shear space. Its complete angular dependence is determined by intensity moments of degrees 0, 2 and 4. In the retained positive-sky compression test, truncation at degree 2 changed the operator by $0.00654243536\,\rho_0$ in operator norm. Retaining degree 4 reduced that discrepancy to $3.34 \times 10^{-16}\,\rho_0$ at the primary quadrature, although higher angular information remained unresolved [E3]. Moment completeness is therefore relative to the response being computed.

Polarization introduces a second response to preserve. For cold-electron Thomson gain in a fixed electron frame, let $\mathsf{F}(n)$ be the transverse polarization matrix, $P_n = I - nn^T$, and $M = \int \mathsf{F}(n)\, d\Omega$. The gain is

$$\mathcal{G}\mathsf{F}(n) = \frac{3}{8\pi} P_n M P_n.$$

Its finite Hermitian carrier determines the gain, while the full collision operator also retains the incident-field loss. The positive intensity fields $I_\pm = 1 \pm 0.6 P_4(n_z)$ have identical gain but different shear operators. Their global intensity lower bound is 0.4, and the computed operator separation is 0.5744626567 in the example's normalization [E4]. A compression sufficient for gain is consequently insufficient for the joint gain-and-shear question.

Near the positive unpolarized reference, the joint linear carrier can be written

$$x_{\mathrm{rad}} = (\rho, \pi, H_4, P_*, A_*), \qquad \dim x_{\mathrm{rad}} = 1 + 5 + 9 + 5 + 3 = 23.$$

Here P_* is the integrated linear-polarization STF contribution and A_* the circular sector. Excluding a circular target leaves dimension 20. Exact rational angular-moment checks independently confirmed these ranks [E4]. They specify what to preserve for these outputs, rather than requiring the full angular field for every subsequent question.

The same calculation makes the physical consequence of polarization omission explicit:

$$C_\pi = -\gamma_T \left(\frac{9}{10}\pi + \frac{1}{5} P_* \right), \qquad \Sigma_{\mathrm{true}} - \Sigma_{\mathrm{unpol}} = -\frac{\gamma_T}{5} \mathcal{L}_I^{-1} P_*.$$

For a positive invertible $\mathcal{L}_I$, this gives $\|\Delta\Sigma\| \le \gamma_T \|P_*\| / [5\lambda_{\min}(\mathcal{L}_I)]$. The finite positive-screen example attained a bias and bound of 0.0072 in its declared toy time units [E4]. The measurable radiation moments specify the operator; the transport derivative/collision residual specifies the right-hand side of the conditional shear inverse. This is the precise place at which a physical history assumption enters.

The next refinement is a directional shear-support plot against the supplied residual allowance, comparing full and reduced responses on identical T/Q/U inputs. The existing result is already a complete conditional response calculation. A Planck application must add the measured channel, spectral, beam/mask and joint-error definitions before giving that plot an observational interpretation.

2. Kinematic quantities from optical and transverse responses

For one specified smooth congruence, the local longitudinal expansion is

$$\mathcal{H}_u(n) = H - \frac{A \cdot n}{c} + \sigma : nn = H(1 + d \cdot n + S : nn), \quad d = -\frac{A}{cH}, \quad S = \frac{\sigma}{H}.$$

The antisymmetric velocity gradient drops out of this contraction. Nevertheless, nonlinear optical observables retain informative couplings. With $\kappa = 5/\ln 10$, the intercept-subtracted magnitude field is $-\kappa \log(1 + d \cdot n + S : nn)$. Its STF octupole obeys

$$O_\mu = \kappa L_S d + R_3, \quad L_S d = \mathrm{STF}(d \otimes S), \quad L_S^T L_S = \frac{S : S}{3} I + \frac{2}{5} S^2.$$

The direct dipole, quadrupole and octupole therefore provide a joint compatibility test of a physical response. The retained source-level calculation pushes the full covariance through $[-\kappa L_S, I_7]$ and uses one remainder budget for all seven octupole coordinates. A finite counterexample has perpendicular and parallel normalized residuals of 0.8 each: both separate checks pass, while their combined norm is $\sqrt{1.28} = 1.13137$ and fails the common unit budget [E5]. The test constrains the stated response/error class; its failure does not identify one unique astrophysical cause.

Transverse information supplies a complementary projection. For a spatial gradient K, define $H(n) = n^T K n$ and $Q(n) = P_n K n$. Then

$$K = 3 \left\langle [nH(n) + Q(n)] n^T \right\rangle_{S^2}.$$

In the ideal Hermitian two-channel extension, combining a full internal detector span with radial and transverse responses identifies 36 real gradient components. Free common catalogue spin removes three geometric vorticity directions, leaving rank 33. These are calibrated response ranks; a polarimetric flux measurement supplies a gradient component only through its actual source/detector map [E6].

The selected Gaia field gives an observationally anchored example of this separation. The retained normal equations encode 755,850 selected rows and all cross terms between the common electric quadrupole and nuisance coefficients. Under the reported marginal proper-motion errors and stipulated cross-source independence, both the six-common-spin/glide fit and the all-97-nuisance fit retain five electric-quadrupole directions. Their signed STF coefficients, in $\mu\text{as}\,\text{yr}^{-1}$, are [E7]:

Nuisance treatment	STF 1	STF 2	STF 3	STF 4	STF 5
Six common spin/glide terms	-2.2624	2.2718	-3.6530	-2.7187	2.3952
All 97 retained nuisance terms	-5.7272	1.1566	-7.4275	-5.5617	6.1770

The two rows describe the same selected data under different nuisance projections. Their change is a direct demonstration of the role of the shared field and covariance. The primary working-law 80%-power radii range from 1.60963 to 2.29431 $\mu\text{as}\,\text{yr}^{-1}$ along its information eigenvectors [E7]. They quantify the design's sensitivity to a common response tensor, not a calibrated matter-shear measurement.

For comoving sources and observer in the declared weak fixed-axis homogeneous geometry, that response is

$$\mu(n) = P_n \mathcal{A} n, \qquad \mathcal{A} = \frac{2c(B_o - B_s)}{D} - \dot{B}_o.$$

The nearby limit gives $\mathcal{A} \simeq \sigma_o$, while a negligible endpoint term gives $-\sigma_o$. The measured sign therefore has history-dependent meaning. In the common-distance parameterization $\mathcal{A} = kq - s$, the ten endpoint/derivative coordinates have a five-dimensional observable image and the null direction $\delta s = k\delta q$. Keeping both endpoint and derivative quantities makes that degeneracy explicit rather than silently assigning the same angular field to one physical tensor.

The next refinement propagates a supplied finite-history/optical remainder through these retained normal equations and displays the resulting physical image. It reuses the measured signed field and full cross blocks without inventing new source-dependent depth moments.

3. Physical-family comparison and geometric stability

Kinematics contributes to geometry through field equations and constraints. For a hypersurface-normal congruence, write $h = H/c$, $S = \sigma/c$, and $\kappa_g = 8\pi G/c^4$. The relevant Gauss relations are

$$^3R = 2\kappa_g \rho + 2\Lambda - 6h^2 + S : S,$$

$$^3R_{\langle ij \rangle} = E_{ij} + \frac{\kappa_g}{2}\pi_{ij} - hS_{ij} + \text{STF}(S^2)_{ij}.$$

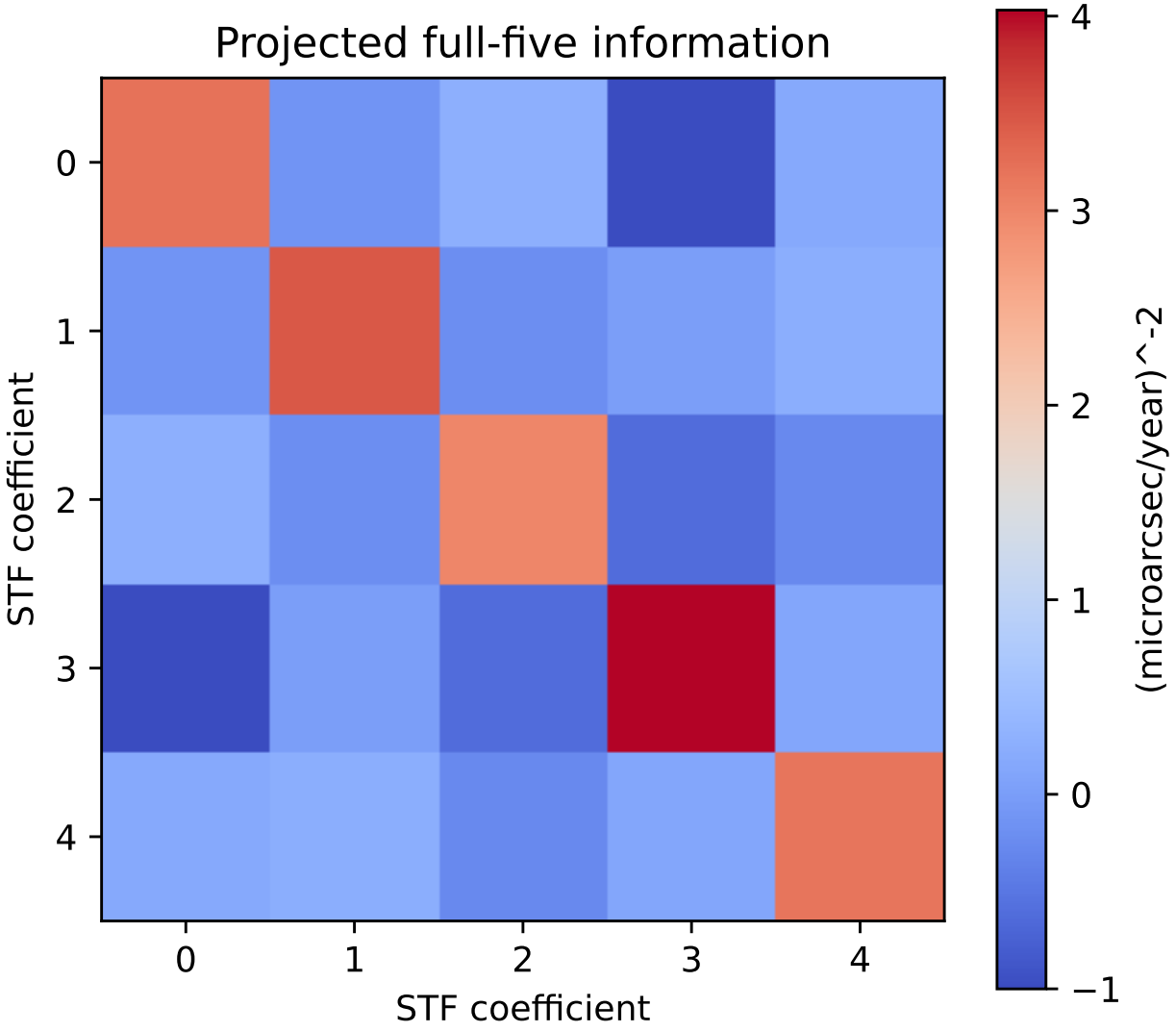


Figure 3: Selected-sample proper-motion information for the common STF response, with nuisance dependence retained.

The electric-Weyl and anisotropic-stress terms show what additional physics is required to turn a shear image into a curvature image. A tensor classification can then compare supplied physical families after orientation, nuisance and approximation error have been included.

The retained type-II and unequal-VII0 finite witnesses have sorted Ricci spectra $(-1/2, -1/2, 1/2)u_*^2$ and $(-3/2, -1/2, 3/2)u_*^2$. Their scalar curvatures are both $-u_*^2/2$, but their rotation-orbit separation is $\sqrt{2}u_*^2$. Their quadratic curvature invariants are $3u_*^4/4$ and $19u_*^4/4$, so a suitable scalar invariant also distinguishes this noiseless pair [E8]. The physical advantage of retaining the tensor is the controlled joint response and uncertainty calculation, not an assertion that every scalar is inadequate.

For family images $\mathcal{M}_k$, a common confidence ball of radius r_α and physical approximation radii ϵ_k gives

$$\widehat{\mathcal{K}}_\alpha = \{k : \mathrm{dist}(z, \mathcal{M}_k) \le r_\alpha + \epsilon_k\}.$$

The sufficient separation condition is $\mathrm{dist}(\mathcal{M}_k, \mathcal{M}_l) > 2r_\alpha + \epsilon_k + \epsilon_l$. In the executed Gaussian geometry-response example, the common likelihood-ratio carrier has five STF coordinates; a sixth trace direction remains useful for detecting that both listed families are inadequate [E8]. This complements classification with a test of the family list itself. Near the FLRW limit, overlapping physical images naturally yield multiple compatible labels.

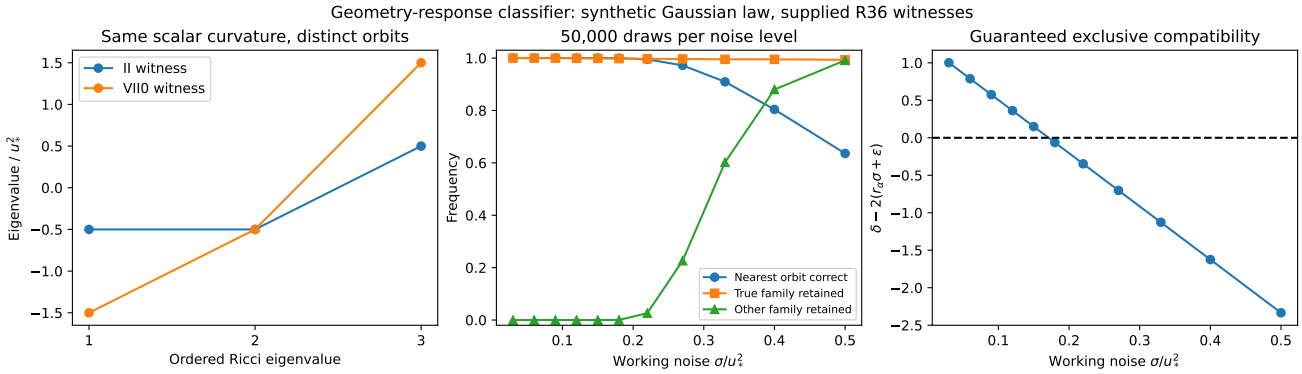


Figure 4: Finite geometry-response classification: equal scalar curvature, separated tensor orbits, and compatibility as the error budget changes.

Morphological stability has an exact local formulation as well. For fixed simple-spectrum quadrupole Q, let $v = O : Q$, $t = Q : Q$, and $m_i = t/3 + 2\lambda_i^2/5$ in its eigenbasis. The distance in ambient STF3 Frobenius norm to a zero-chirality surface is

$$\rho_{\mathrm{sign}} = \min_i \frac{|v_i|}{\sqrt{m_i}}.$$

The retained sharp witness attains $1/\sqrt{3}$; an independently constructed boundary perturbation makes chirality vanish to numerical precision [E8]. This turns a chosen perturbation-radius test into a radius tied to the actual sign-changing geometry. Uncertainty in Q requires its own joint image rather than substituting a point estimate into the fixed-Q theorem.

The next refinement is a common-confidence comparison of already supplied geometry or PM-history families, including the adequacy residual. CMB Bianchi labels enter through full, source-qualified temperature/polarization family responses with their histories and processing conventions.

4. Directional expansion and optical transmission

The released supernova magnitude field already supplies a concrete signed angular result. The primary Pantheon+ sample excludes calibrators and uses $0.0233 \leq z_{\rm HD} \leq 0.15$, split at 0.06. A global intercept and one depth offset are profiled. Dipole and STF-quadrupole terms in each bin give a 16-coordinate angular carrier. The released STAT+SYS covariance is used once, retaining cross-depth terms [E9].

Cohort	Low/high rows	Isotropic residual statistic	Angular residual statistic	Improvement	Conditional upper-tail probability
Primary	342 / 148	433.4585	416.2519	17.2066	0.37235
Lower-redshift sensitivity	557 / 148	608.5020	591.9479	16.5541	0.41500

Under the stipulated fixed-design Gaussian law, the angular improvement is below the 5% χ^2_{16} threshold of 26.2962 in both overlapping cohorts. The primary dipoles are $(0.01758, 0.03002, 0.01848)$ mag at low depth and $(0.00223, 0.02426, 0.05879)$ mag at high depth. The full signed vector and its covariance retain how these angular patterns change together [E9]. These are apparent-magnitude responses; the profiled intercept does not provide an absolute H_0.

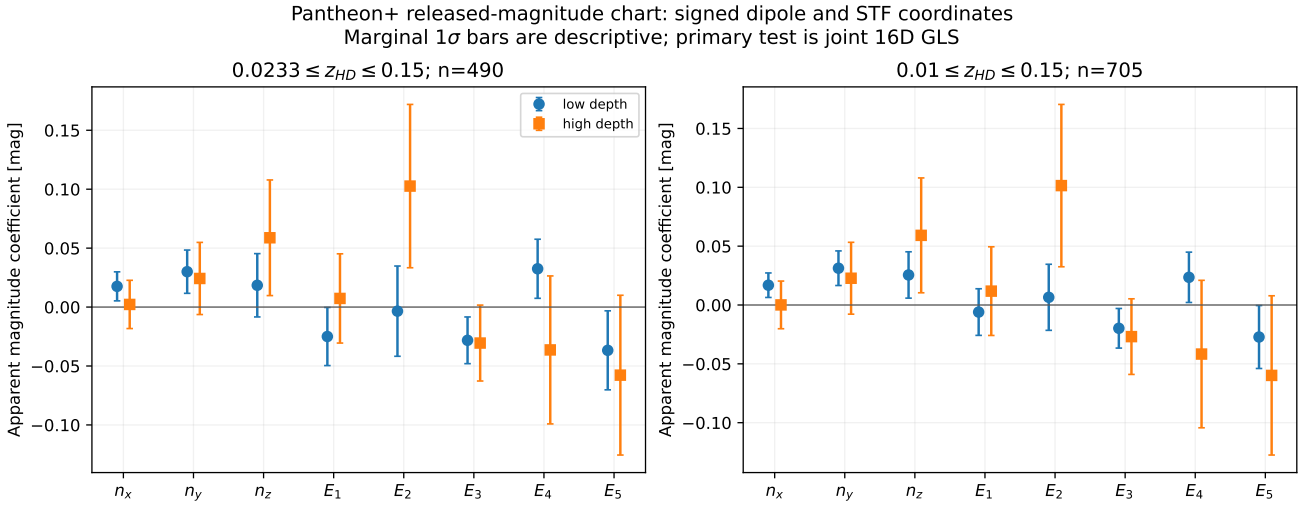


Figure 5: Released-magnitude dipole and quadrupole coefficients in two depth bins; displayed marginal errors accompany the retained full covariance.

The consumed covariance subblocks are symmetric and positive definite. The full release array fails the frozen symmetry tolerance outside their support, so its validation does not extend beyond the selected blocks. The primary cross-depth block contains 50,616 nonzero entries [E9]. Treating the bins as independent would discard a measured part of the adopted release covariance.

CF4 illustrates a different uncertainty mechanism. The exact retained H03 operator acts on 35,970 selected groups. A stipulated common Gaussian velocity field over $0.005 \leq k \leq 0.05\,h\,\mathrm{Mpc}^{-1}$, with the specified independent source and method-offset terms, yields a conditional covariance trace of 17.68057 $(\mathrm{km\,s^{-1}\,Mpc^{-1}})^2$ for the 36-coordinate $R = 5\,h^{-1}$Mpc result [E10]. This transports a common physical field through the actual selected operator. It does not add the inherited working variance again or turn an assumed velocity law into an empirically calibrated selection model.

Optical transmission enters the physical interpretation through

$$D_{L,r}(z,n) = \frac{cz}{\mathcal{H}_u(n)} \exp\left[r_g(z,n) + \frac{1}{2}\tau_r(z,n)\right].$$

If $|r_g| \le b_g$ and $\tau_- \le \tau_r \le \tau_+$, the flux-based $H_F = cz/D_{L,r}$ gives

$$H_F e^{-b_g+\tau_-/2} \le \mathcal{H}_u(n) \le H_F e^{b_g+\tau_+/2}.$$

This interval keeps finite-distance and attenuation assumptions visible. For a calibrated matrix channel $\mathcal{A} = D_L^{-2}\mathcal{P}$, its input effect $\mathcal{A}^*(I)$ constrains distance; relative channel shape does not determine grey transmission.

For distance $d_r = de^{\tau/2}$, the scalar CBL relation has the exact correction

$$C[d_r] = 1 + e^{\tau}(C[d] - 1) + \frac{d_r^2}{2c^2}(H^2\tau'' + HH'\tau').$$

In the flat synthetic control with $\tau = 0.2z$, the uncorrected residual at $z = 1.5$ is -0.3498588; exact correction returns zero to roundoff. The declared opacity class instead gives the outer geometric interval $[-0.3802306, 0.0225]$ [E11]. This demonstrates how a flux residual can coexist with flat geometry.

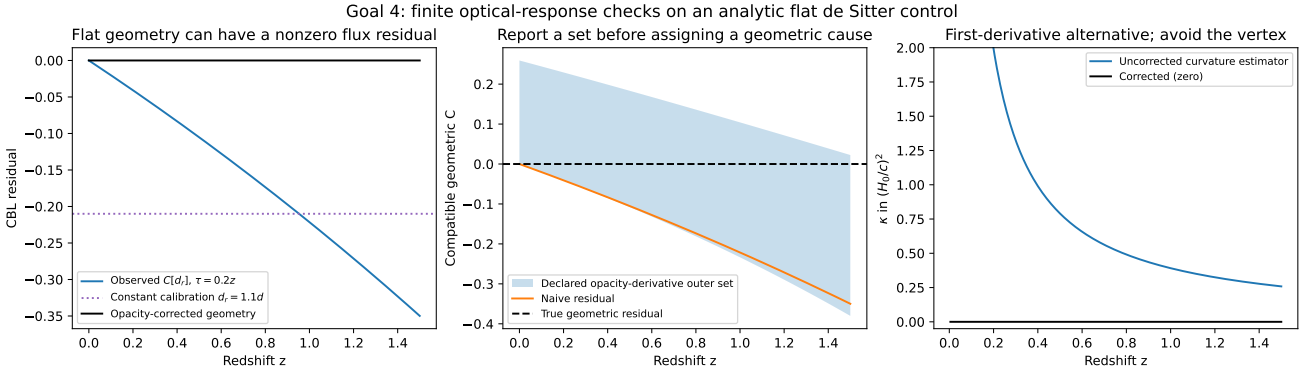


Figure 6: Flat-geometry optical control: attenuation-induced CBL residual, its corrected value, and the permitted geometric interval.

The next empirical refinement uses a calibrated distance function and independently qualified $H(z)$ to test constancy of $\kappa(z) = [H^2 d'^2/c^2 - 1]/d^2$, carrying opacity and calibration through one joint event. The existing angular SN carrier supplies the directional observation component; it does not by itself supply absolute distance calibration or redshift derivatives.

Evidence map and reusable source definitions

Source commit: `bdead6a8f51282ca647224b460b0f65de085568b`. The accompanying `BASELINE_REPORT.json` records exact repository paths, numerical selectors, source hashes, figure bindings and scoped qualification for each entry. No catalogue was refit for this report.

Evidence	Bound result
E1	Saved Gaussian compression and common-family likelihood comparison
E2	Joint physical remainder and fixed-law coverage experiment
E3	R36 radiation response, target coverage and moment truncation
E4	Joint Thomson/shear carrier and polarized collision omission
E5	R36 source-level optical inverse and shared octupole budget
E6	Detector/angular rank and frame-spin quotient
E7	R36 selected-sample electric-quadrupole history response
E8	R36 spatial Ricci witnesses and fixed-Q chirality radius
E9	R36 released SN angular coefficients and selected covariance
E10	R36 correlated CF4 covariance through the source-bound operator
E11	Exact opacity correction and finite CBL control
E12	Finite fibre-count information and calibration experiment

The physical response conventions follow the primary derivations identified in the retained source notes: Challinor's covariant polarized multipoles, Marcori et al.'s observer/source proper-motion response, the Gauss relations in relativistic cosmology, and the Clarkson–Bassett–Lu distance/expansion consistency relation. Their application here is specified by the source frames, response maps and uncertainty laws above.

- A. Challinor, Microwave background polarization in cosmological models, Phys. Rev. D 62, 043004 (2000), astro-ph/9911481.
- O. H. Marcori et al., Direction and redshift drifts for general observers and their applications in cosmology, Phys. Rev. D 98, 023517 (2018), 1805.12121.
- C. G. Tsagas, A. Challinor and R. Maartens, Relativistic cosmology and large-scale structure, Physics Reports 465, 61–147 (2008), 0705.4397.
- C. Clarkson, B. A. Bassett and T. H.-C. Lu, A general test of the Copernican Principle, Phys. Rev. Lett. 101, 011301 (2008), 0712.3457.